



Faculty of Physical Education for Girls
Department of Athletic Training and Movement Sciences

**"The effectiveness of technical performance endurance exercises on
the swimming time of 400 m freestyle for junior swimmers"**

Research Submitted by

Adam Ismail Mohamed Ali

Swimming Coach At the Egyptian Olympic club
Researcher at the Department of Athletic Training and Movement Sciences

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Under the auspicious of

Dr.

Galila Hassan Mohamed Ibrahim

Professor of Athletic training (water sports) at the
Department of Athletic Training and Movement
Sciences

Faculty of Physical Education for Girls Alexandria
University

Dr.

Naglaa Mohamed Ahmed Shakra

Teacher at the Department of Athletic Training
and Movement Sciences

Faculty of Physical Education for Girls Alexandria
University

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- **Research Introduction and Problem:**

The physical abilities with its various elements such as endurance , speed, strength, agility and flexibility are of great importance in the preparation of the individual and developing his fitness of the type of activity at high technical, level and in an appropriate way taking advantage of the energy of the individual and all his forces in the right direction to achieve the best results, To improve the level of physical abilities and its various components.

Mohamed Hassan Allawi (1994) states that physical abilities give the athlete the basic physical qualities in a comprehensive and balanced manner, which is the cornerstone of all sports practitioners. (62) (35: 35)

Elsayed Abdel-Maqsoud (1997), Essam El-Din Abdel-Khalik Moustafa (2005), refer that endurance is one of the most important physical qualities that help to continue the effort efficiently. The level of endurance is related to the appearance of fatigue. If an individual performs an effort with a specific load, after a period of continuous or intermittent performance finds difficulty to perform. (29: 145) (17: 67)

Aboelela Ahmed Abdel fatah, Hazem Hussien Salem (2011) confirm that the most important elements of the physical abilities of swimming is endurance as it is considered the key to success which leads to reaching the upper levels in swimming. The construction of a wide base of endurance affects the ability to achieve high speed rate in swimming during competitions and training. (1: 117)

Maglisco (1993) adds that endurance training aims to improve aerobic ability so that swimmers can swim faster with less dependence on anaerobic representation. And thus Lactic acid will build up at a lower volume and postpone fatigue. Swimmers who perform endurance training at a faster rate in three quarters of most races and will have the energy to increase speed to the end. Endurance exercises are important for swimmers in all competitive distances from 100 meters up to longer distances the importance of endurance exercises increase the longer the race (57: 80, 81)

Fahmi Al-Beek and others (1996) mention that when training junior swimmers, attention must be paid to developing aerobic endurance and improving the performance of the four swimming paths before resorting to anaerobic training. The importance of aerobic endurance training for training junior swimmers is shown to help to juniors learn and master the proper swimming, And increases the ability of the junior when performing aerobic endurance exercises to master both movements of the arms and blows of the two legs, and helps to show performance errors clearly from the anaerobic training, which helps to correct them, and increases the confidence of the junior when he can cover a long distance at a regular speed. (49: 255)

Aboelela Ahmed Abdel fatah (1998) and Mufti Ibrahim Hammad (1998) mention in The field of training in general and swimming in particular, agree



importance of developing (aerobic – anaerobic) endurance as an essential basis for achieving what is known as performance endurance which . High-intensity physical and skills abilities for as long as possible, so that the player needs aerobic endurance to achieve the efficiency through out the race. (4: 269, 270) (46: 142)

In light of this, Abul-Ela Ahmed Abdel-Fattah (1997) points out that endurance is the ability to continue effectively in performing a physical work that has a positive effect on the processes of building the components of the specialized sports activity as a result of raising the level of adaptation to the performance of physical loads and transferring their influence to the specialized activity. (3: 162)

Essam Mohamed Helmy (1997) and Mohammed Ali Al-kot (2005) believe that the goal of endurance training is to help the swimmer obtain the speed rate of the step during the middle of the race without showing any symptoms of fatigue during the middle of the race , the contribution of endurance training show in light of the race distance 42: 164) (31: 163)

" Aboelela Ahmed Abdel fatah, Hazem Hussien Salem (2011) point out that the most important things to consider when planning training for junior swimmers is to pay attention to the technology and proper performance of the four swimming styles as . If the correct performance is not taught in the young age stages, it becomes difficult to develop, Performance level In the following stages. The development of aerobic endurance is one of the most important factors that should be developed in the junior stages because it is the basis on which the requirements of the achievement of the higher levels are adopted, as the juniors swimmer is more capable of aerobic training and learning motor skills he also pointed out that LTAD refers to the basics stage, teaching training, training for training, training for competition, training for winning. The Swimming Federation of the United States of America has designed a program to develop the level of swimmers according to a national program called the various challenge program. This program is designed for swimmers under the age of 12 and includes 5 races, 200 m freestyle - 100 m back stroke - 100 m breast storke - 100 m butterfly - 200 m mixed. (117-275)

Maglischo (2003) points out that in order to achieve the highest athletic achievements for junior swimmers, we must work with the thought of the third millennium. The system of the advanced world (LTAD) the swimmer should not focus only on specialization in swimming but according to modern trends should exercise other sports beside swimming, which ensures balanced physical development and reduces boredom. The general rule of specialization is: Until the age of 12 years, the child is a swimmer specialized in 200 meter race, up to 14 years of age. The child is a 400 meter mixed swimmer. (56: 401)

Abu al-Ela Ahmed Abdel-Fattah, Mohamed Sobhy Hassanein (1997) agrees that one of the major problems in swimming is that most coaches are interested in developing speed by increasing the effort, especially in the early stages of swimming.



the skill performance which is the base of speed . Through the biomechanical analysis of champions in major tournaments found that the fastest swimmer is not always the winner. Sometimes he loses the race because of some basic skills. When the owners of the world numbers were asked how they felt when they scored, they all agreed that they recorded it with minimal effort, and they did not expect the record compared to the effort. (4: 54)

Osama KamelRateb, Mohamed SobhyHassanein (1998) confirms that in order to achieve athletic excellence in the field of swimming training, the components of training loads should be upgraded and attention should be paid to the first element of fitness in swimming, which is endurance, as evidenced by the analysis of previous studies conducted in Sydney Australia 2000 They found that most swimmers who achieved the highest records did not rely on their speed training but relied on performance, development and relating it to load through the distribution of effort - reducing the resistance - the flow and sway - the economy of effort and the number of strikes. (6: 117)

As confirmed by many scientific references and previous studies, including Fahmi al-Beek and others (1996), Mohamed Ali al-kot (2005), RawdaHamdy Ibrahim (2011), Khalid Hassan Tawfiq, Ahmed Jaber Abdul Rashid (2012) on the importance of building this phase in the light of improving Performance in terms of endurance as the basic element in fitness for developing any swimmer and maintains the performance for a long time without feeling tired as well as avoiding the wrong performance and common mistakes in swimming that occur due to fatigue.

Also, through the experience of the researcher as a player and coach, he found that the performance level of swimmers decreases at the end of the race clearly than the beginning which may be due to the concentration of most of the coaches to achieve rapid numbers without taking into account fixing the performance technique which results in excellence in the time of covering the distance at a certain stage and then stopping development the level After that, in the major stages of the world championships, where the swimmers continue to keep at a certain number that they does not go beyond until they become desperate and the swimmers disappear before they start. On the other hand, the training method used for the younger stages depends on the speed and neglect of the first element, which is endurance.

Research Objective:

The aim of the research is to improve the swimming time (400 meters) freestyle for junior swimmers through:

1. Design a training program to improve the performance of technical performance.
2. Identify the impact of performance training exercises on capacity improvement and endurance measurements for junior swimmers.

3. Identifying the effect of performance endurance exercises on 400 m freestyle time.
4. Identifying the effect of performance endurance exercises on improving the skill level of 50m freestyle swimming.

Research hypotheses:

- 1- There are statistically significant differences in the tolerance measurements of the two groups (experimental - control) for the swimmers that are emerging between the two measurements (the tribal - the dimension) in favor of the experimental group.
- 2 - There are statistically significant differences in the swimming time of 400 meters free for the two groups (experimental - control) for swimmers emerging between the measurements (tribal - remote) for the benefit of the experimental group.
- 3- There are statistically significant differences in the skill level of the 50m freestyle swimming for the two groups (experimental - control) for the swimmers who are born between the two measurements (tribal - remote) in favor of the experimental group.

Research procedures:

Research population:

The researcher used the experimental method using (equal groups of two groups experimental - control) as it suits the nature of the research.

The research sample:

The researcher selected the research sample by the deliberate manner from juniors (32) swimmers at 11 years stage from the Egyptian Olympic Club in Alexandria Governorate. The swimmers were divided into (20) swimmers for the basic research sample and (12) swimmers for the exploratory study.

Equipment and tools used:

- 1 - Medical balance to measure weight.
- 2 - Ratamir to measure length.
- 3- Floating plate for the strikes of the two legs.
- 4 -Buoy to fix the two legs.
- 5- Small fins to do the work of the blows of the two legs.
- 6- Big fins to withstand the blows of the two legs.
- 7- Stop watch to measure the performance time.

Tests and measurements used:

- Physical fitness tests related to load.
- Test of the level of record achievement (50 m), (400 m) free.
- Test of the level of skill performance of 50m freestyle swimming.

The exploratory Study:

The first exploratory study:

The researcher conducted the first exploratory study in the period from (12/8/2016 to 20/8/2016) in the Olympic Club swimming pool.

The second exploratory study:

The researcher conducted the second exploratory study in the period from (5/9/2016 to 7/9/2016) on the basic research sample (experimental group).

Procedures for applying the research experience:

Pre measurements:

The pre measurements were conducted from (28/9/2016 to 30/9/2016) in the swimming pool and fitness hall of the Egyptian Olympic Club in Alexandria.

Application of the training program:

The training program was implemented in the period from 1/10/2016 to 7/2/2017 in the swimming pool of the Egyptian Olympic Club in Alexandria.

Post measurements:

The researcher conducted the post measurements of the proposed training program from (8/2/2017 to 10/2/2017)

Statistical processsors:

1. Arithmetic mean.
2. Standard deviation.
3. Median.
4. Torsion coefficient.
5. Pearson correlation coefficient.
- 6 .T test for statistical significant differences



7. Variance Analysis test.
8. The percentage of improvement between median.

I. Conclusions:

In light of the objectives and the limitations of the research and within the nature of the methodology used and the sample to which the research applied to and tests used and through the discussion and analysis of the findings reached by the researcher the following was concluded:

- The proposed training program using performance endurance exercises has a positive effect on the improvement of physical abilities related to load.
- The proposed training program using performance endurance exercises has a positive effect on the improvement of the record level of 400 meters free style swimming.
- The proposed training program using performance endurance exercises has a positive effect on the improvement of the skill level of the 50m freestyle swimming.

Recommendations:

In light of the study conducted by the researcher and the results reached , and based on the research methodology, the proposed program and the sample to which the study was applied, the following recommendations could be reached:

- Paying attention to the element of endurance because it is the basis for the development of any swimmers, especially the juniors sector.
- The need to pay attention to standardizing the loads used according to the abilities of juniors.
- The need to pay attention to the period of establishment of endurance at the beginning of the season by not less than 7 weeks.
- Conducting further studies and research on other methods of swimming.

